

YOLO11-Occ: An Improved Occlusion-Robust Object Detection Method Based on YOLO11

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Abstract—Object detection is widely applied in various fields, including intelligent surveillance, intelligent urban systems, and environmental perception for Advanced Driver Assistance Systems (ADAS). However, in road scenes, the presence of numerous objects and mutual occlusions leads to challenges in feature extraction and reduces detection accuracy for occluded objects. This paper proposes an occlusion-aware object detection algorithm based on YOLO11 to address these issues. Specifically, the input image is segmented into superpixels, and corresponding reactors are generated for each superpixel. A KNN-based adjacency graph is then constructed to extract graph features. Furthermore, we introduce an occlusion attention mechanism to guide the superpixel graph in focusing on occluded regions and computing occlusion loss. Our approach is evaluated on the KITTI dataset. Experimental results demonstrate that, compared to the state-of-the-art detection algorithm YOLO11, our method achieves a 10.28% improvement in mAP50, effectively enhancing detection performance under occlusion scenarios.

Keywords—object detection, graph neural networks, reactor, occlusion awareness

I. INTRODUCTION

Autonomous vehicles (AVs) are revolutionizing transportation, promising safer and more efficient mobility solutions [1]. A fundamental aspect of AVs' functionality lies in their perception systems, which rely on an array of sensors such as cameras, LiDAR, and radar to construct a detailed and accurate representation of their surroundings. Despite significant advances in sensor technology, the challenge of occlusions—where objects partially or fully block each other—remains a critical barrier to achieving reliable and robust perception, particularly in complex urban environments.

Occlusion frequently occurs when vehicles are obstructed by other vehicles, buildings, or pedestrians, creating significant difficulties in detecting these targets on time. This issue not only impacts the accuracy of object detection algorithms but also poses substantial risks to traffic safety, leading to potential accidents and compromising overall traffic efficiency. Thus, addressing the occlusion problem is essential for improving the reliability of AV systems and ensuring the safety of both vehicles and pedestrians.

Recent developments in the field of object detection have sought to address these challenges through deep learning-based approaches [2]. However, existing detection methods often struggle in occluded scenarios, resulting in missed detections or reduced accuracy. In response to this limitation, innovative techniques that focus on enhancing the detection of occluded objects are being actively explored. Some approaches integrate multiple sensors or employ specialized algorithms to handle occlusions more effectively [3], but these

methods still face challenges in achieving both high detection performance and low computational complexity.

This paper proposes an advanced object detection framework to handle occlusion in autonomous driving scenarios. By leveraging a multi-sensor fusion approach and incorporating occlusion-aware mechanisms, our method improves the detection of vehicles and pedestrians in highly occluded environments. The key contributions of this work include:

- We introduce a fusion module that can fuse the superpixel mask position coding with the image feature.
- We propose a method based on superpixel graph convolution, which can use the adjacency relationship between the target and the occluder to expand and complete the occluded features.
- We present an occlusion attention mechanism that uses a self-attention mechanism together with a convolution operation to enhance the features of the occluded part.

II. RELATED WORKS

A. Superpixel

Ren et al. [4] first proposed the concept of superpixels. According to different label assignment modes, the existing superpixel segmentation methods can be classified into clustering-based methods and deep-network-based methods. Our approach selects the clustering-based method.

Clustering-based methods iteratively refine the labels of pixels based on the feature similarity until certain convergence conditions are met. In particular, Achanta et al. [5] proposed a simple linear iterative clustering (SLIC) algorithm, which is the most widely-used and prominent model. Li et al. [6] mapped each image pixel to a ten-dimension feature space, and proposed a superpixel segmentation based on linear spectral clustering (LSC). Nagar et al. [7] used reflection symmetry to optimize the size and shape of superpixels (SymmSLIC), Wu et al. [8] proposed a fuzzy SLIC (FSLIC) based on neighboring spatial information and onion peeling.

Deep-network-based methods utilize deep neural network to capture local image features and build associations between pixels and surrounding grids. For example, Jampani et al. [9] generated superpixels (SSN) by fusing deep neural network and differentiable SLIC. It is the first end-to-end trainable network for superpixel segmentation. Pan et al. [10] proposed a fast superpixel generation algorithm that can generate superpixels with lattice topology (LT). Yang et al. [11] integrated convolutional neural network (CNN) and fully convolutional network to generate superpixels on a regular grid.

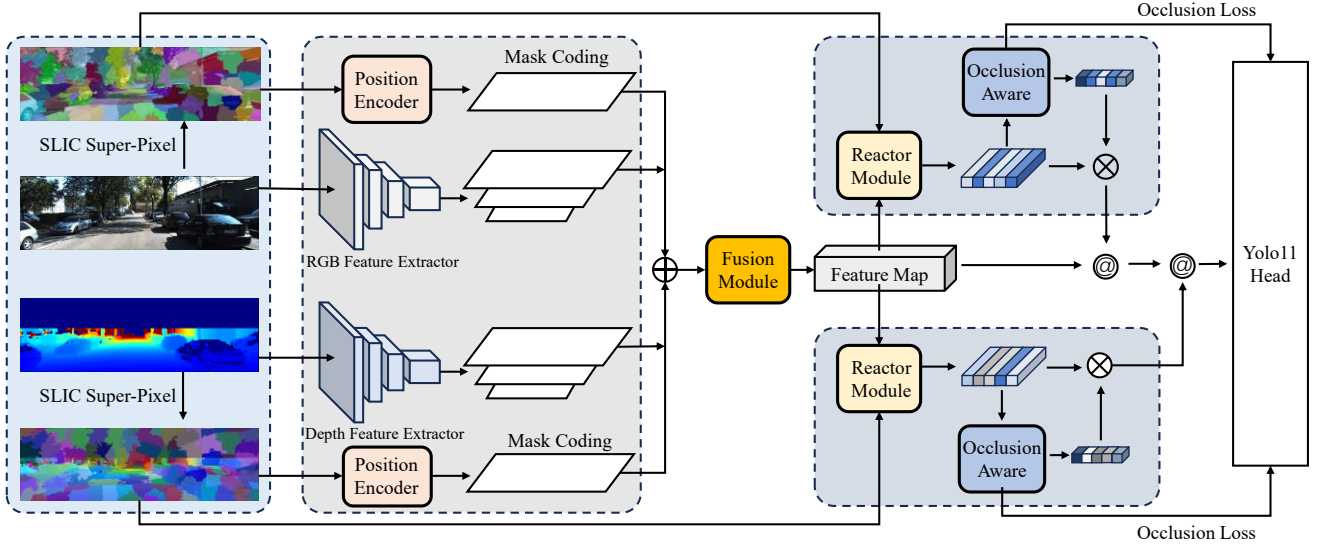


Fig. 1. The overview of YOLO11-Occ. $\otimes$ represents element-wise multiplication, $\oplus$ illustrates element-wise add, and $@$ means convolution operator which select input tensor as kernel.

B. Object Occlusion

Detection. Sheng et al. [12] propose a boosted cascade framework for detecting partially visible objects. However, their approach uses handcrafted features and can only be applied to images where objects are artificially occluded by cutting out image patches. Additionally, a number of deep learning approaches have been proposed for detecting occluded objects [13, 14]; however, these methods require detailed part-level annotations to reconstruct the occluded objects. Xiang and Savarese [15] propose to use 3D models and to treat occlusion as a multilabel classification task. However, in real-world scenarios, modeling occluders in 3D is challenging because their classes are often unknown a priori. Also, other approaches are based on videos or stereo images [16, 17], however, we focus on object detection in still images. Most related to our work is part-based voting approaches [18, 19] that have proven to work reliably for semantic part detection under occlusion. However, these methods assume a fixed-size bounding box which limits their applicability in the context of object detection.

III. METHODS

This research proposes an occlusion-robust object detection optimization method based on a superpixel graph and occlusion attention mechanism. The method consists of four components: feature encoder, feature fusion module, reactor module, and occlusion attention module. The object detection head can be flexibly selected, and this study employs the YOLO11 [20] head. The overall model architecture is illustrated in Fig. 1.

A. Framework Overview

In the feature extraction stage, the encoder module is used to extract the image input I_{rgb} , I_{depth} features f_r , f_d , and encode the position information of the superpixel mask S_{rgb} , S_{depth} to obtain M_r , M_d . In the feature fusion module, the image features and superpixel mask position encoding are merged, the superpixel position information is introduced into the features in the channel dimension and spatial dimension, and a channel transformation convolution is applied to reduce the channel dimensionality and obtain FM . Then, S_{rgb} and FM are input into the reactor module, S_{rgb} is one-hot encoded to SM , and element-wise multiplication, pooling and other operations are used to generate each superpixel reactor R to characterize the saliency features of each superpixel. Next, this method designs a superpixel graph convolutional neural network to extract the features of each neighboring reactor and obtain R_{graph} . In addition, through the occlusion attention module, we obtain R_{attn} and $Mask_w$. The former is convolved as kernel with FM to obtain FM_{attn} that incorporates graph features and occlusion attention, and the latter is convolved with $Mask_{gt}$ to calculate occlusion loss.

In the output head, we use the detection head of YOLO11, take FM_{attn} as input, and obtain the final 2D object detection box.

B. Units

Fusion Module. Its detailed structure is shown in Fig. 2. The image feature f_r , depth feature f_d , and superpixel position

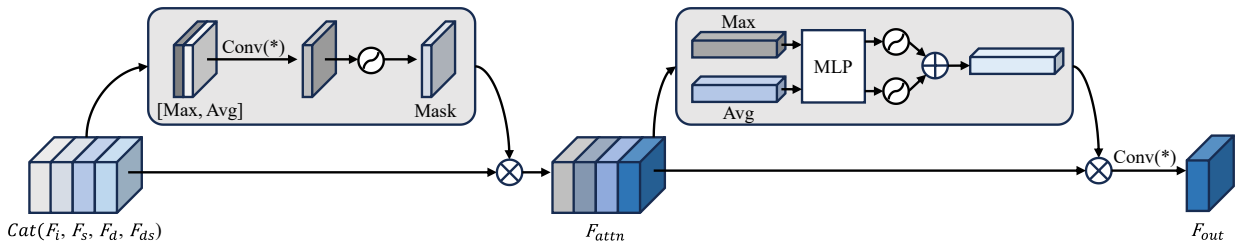


Fig. 2. The structure of the fusion module. $\odot$ means sigmoid function.

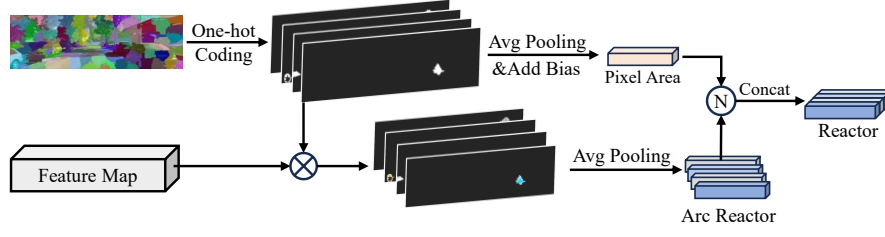


Fig. 3. The structure of the reactor generator. $\odot$ means area normalize operator.

code M_r , M_d are merged in channel dimension. Through channel attention and spatial attention, feature adaptive enhancement is achieved. Subsequently, after channel transformation convolution, the superpixel position information is embedded to FM .

Reactor Generator. The name of reactor comes from the Arc Reactor in Iron Man, which represents the salient core features of superpixels. The structure of this module is shown in Fig. 3.

Taking visible light input as an example, S_{rgb} represents the superpixel segmentation results of the visible light image input, where each pixel $v_{(x,y)} \in [0, n_{spixel}]$ denotes the superpixel label at (x, y) . Subsequently, each superpixel's binary mask is independently extracted using one-hot encoding, yielding SM . This mask is then downsampled to match the size of FM , followed by an element-wise multiplication with FM . A global average pooling operation is applied, and the result is normalized by the superpixel area to obtain the corresponding reactor. The process can be formulated as (1) and (2), where a bias term is introduced to prevent a zero denominator during normalization. In this manner, each superpixel is associated with its corresponding reactor, efficiently enabling the salient feature representation of 2D superpixels.

$$area_i = \text{sum}(Mask_i) + bias \quad (1)$$

$$R_i = \text{AreaNormal}(\text{AvgPool}(Mask_i \otimes FM), area_i) \quad (2)$$

Superpixel Graph Module. The core challenge of the occlusion problem is the lack of information about the occluded part of the target. Graph-related algorithms often explicitly model the relationship between adjacent nodes. This method proposes an information expansion module based on superpixel graphs, whose structure is shown in Fig. 4.

This module consists of a superpixel graph convolution network. We regard each reactor as a node in the superpixel graph. After the fusion module, each node reactor incorporates the adjacency relationship between superpixels after superpixel segmentation and the texture depth information corresponding to the superpixel itself. The negative squared

Euclidean distance (NSD) between each node and other nodes is calculated using the KNN algorithm as shown in (3), where R_i and R_j represent reactors i and j respectively.

$$NSD(i, j) = -\|R_i - R_j\|^2 \quad (3)$$

Then, we select the K nodes closest to the node as the neighboring reactor of the node. The neighbor critical features are concatenated with the features themselves, and conv2d is used to fuse them. Normalize and nonlinear activation function are introduced to obtain the maximum feature value. This process is shown in (4), (5), (6), where $\Phi(R_i)$ represents the K nearest neighbor nodes of R_i .

$$\Phi(R_i) = \arg \text{topk}(NSD(i, j)) \quad (4)$$

$$R_{cat} = \text{Cat}(R_i, \text{repeat}(k), \Phi(R_i) - R_i) \quad (5)$$

$$R_{graph} = \text{Max}(\text{Conv2d}(R_{cat})) \quad (6)$$

Therefore, the saliency features of each superpixel constitute a shared graph, which expands the adjacency information of each reactor and partially completes the information of the occluded area.

Occlusion Attention Module. This research proposes an occlusion attention module. Its structure is shown in Fig. 4. After R_{graph} is calculated by the self-attention mechanism, it is mapped to the interval of 0-1 through the activation layer. The channel is reduced through linear layer to obtain the occlusion attention score S_{attn} of each superpixel. After element-wise multiplying it with R_{graph} , we get the reactor R_{attn} after occlusion attention. SM and S_{attn} are weighted to get $Mask_w$, which represents the possibility mask of each pixel being in an occluded area. This process can be expressed as follows (7), (8), and (9).

$$S_{attn} = \text{Sigmoid}\left(\text{Linear}\left(\text{attn}(R_{graph})\right)\right) \quad (7)$$

$$R_{attn} = S_{attn} \otimes R_{graph} \quad (8)$$

$$Mask_w = \sum_{i=0}^{n_{spixel}} S_{attn}(i) \otimes SM(i) \quad (9)$$

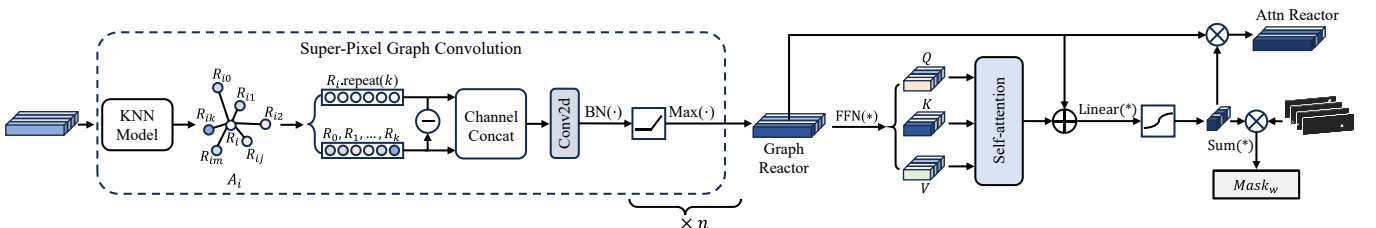


Fig. 4. The structure of the superpixel graph module and occlusion attention module. $\ominus$ means element-wise diff operator.

To further improve the occlusion sensitivity of this module, we introduce an occlusion loss: $Loss_{occ}$. The ground truth GT is initialized to a zero matrix, then, add 1 to each element value in its n_{bbox} ground truth boxes, and normalize it. GT is constructed as shown in Fig. 5. It calculates the binary cross entropy loss of $Mask_w$ and GT .

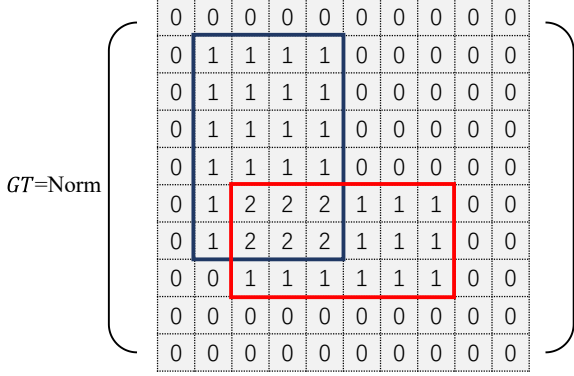


Fig. 5. The production of ground truth.

Finally, R_{attn} is used as the convolution kernel and convolve FM to realize the mapping of the saliency representation in each reactor to its image feature. The convolution results of each kernel are superimposed with the initial FM to obtain the final feature FM_{out} . As shown in Fig. 6.

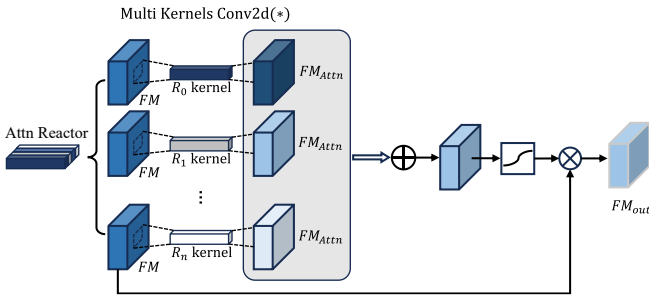


Fig. 6. The details of the output process stage.

IV. EXPERIMENT

This algorithm is based on the KITTI [21] dataset, and a series of comparative experiments are designed to verify the improvement of target detection accuracy between this model and the baseline model. Ablation experiments are designed to test the coupling effect between modules.

A. Experiment Setup

This experiment is based on the KITTI object detection dataset. It is one of the most representative datasets in the field of autonomous driving, covering a variety of perception tasks, including object detection, semantic segmentation, and depth estimation. This study mainly uses its 2D object detection subset, which includes RGB images, LiDAR point clouds, annotation information, and camera calibration data. It contains 7481 training images and their corresponding annotation files. Since the test set is not annotated, we divide the training set into a ratio of 7:3 for training and evaluation.

Our model is adapted based on YOLO11. It is trained on a single NVIDIA Tesla V100 GPU. We use an input resolution of 640×640 . The optimizer employed is SGD with an initial learning rate (lr0) of 0.01 and a final learning rate (lrf) of 0.01. The model is trained for 20 epochs with batch size=32. To facilitate feature fusion, the method in this experiment does not perform feature enhancement.

B. Result

In terms of detection accuracy, the experimental results are shown in the following Table I. We use the YOLO11 model as the baseline and compare the increase in mAP50 accuracy. Compared with the YOLO11 model, the detection accuracy of this algorithm increases by 10.28%, further demonstrating the effectiveness of this algorithm.

TABLE I. RESULT OF DETECTION ACCURACY

Methods	mAP50	mAP50-95
YOLO11	0.7279	0.4697
YOLO11-Occ	0.8307	0.6141

In terms of module coupling, we designed an ablation experiment to test the coupling effect of the fusion module and the graph attention module. The results are shown in the following Table II. Our fusion and occlusion attention modules have greatly contributed to the increase in model accuracy, thus further verifying the coupling relationship and effectiveness of each module.

TABLE II. RESULT OF MODULE COOPERATION

Methods	mAP50	mAP50-95
YOLO11	0.7279	0.4697
YOLO11+F	0.8185	0.5713
YOLO11+F+OA	0.8307	0.6141

The inference results of our algorithm are visualized in Fig. 7. This algorithm can still effectively detect the target when it is occluded, which shows its effectiveness.



Fig. 7. The inference result visualization of our method. Each bounding box is shown with a class label and confidence.

V. CONCLUSION

This paper proposes a target detection algorithm based on superpixel graph convolution and occlusion attention to enhance the detection of occluded targets. The method leverages boundary-sensitive superpixels to better distinguish target features from occluders. A reactor generation module is employed to highlight salient two-dimensional superpixel features, while a superpixel graph is constructed to facilitate superpixel convolution, effectively enriching the feature representation of occluded regions. Finally, an occlusion attention module is applied to further enhance the extracted features. Compared with the YOLO11 baseline, our YOLO11-Occ method achieves a significant 10.28% improvement in mAP50, demonstrating its effectiveness in addressing occlusion challenges. These results highlight the significance of this research, while also indicating ample room for further refinement and optimization in future work.

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